CMP6200

Individual Undergraduate Project (FYP)

Project Interim Report

NeuroScan Nepal: Brain Tumor Detection & Medical Advisory using AI

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1 Introduction and Context

Access to quality healthcare remains a significant challenge in Nepal, particularly in rural and remote regions. The health concerns that affect the brain include tumours and lesions, and these are some of the most important issues from a healthcare perspective. This is due to the fact that they require skilled manpower and specialized technology for the diagnosis of brain disease that cannot be obtained easily throughout the country. Because of this, there are long delays in diagnosing patients with these conditions, and in many instances, when diagnosed, the patients' situation is already untreatable.

Nonetheless, despite the fact that there are 30 million people in Nepal, the total number of neurosurgeons in the country is only around 30 to 40 at present, which makes the ratio of one neurosurgeon per 750,000 people. It means that the proportion is way below the recommended number of one neurosurgeon for every 100,000 people by WHO (Shrestha, 2024; Alokozay et al., 2025). The centers where MRIs are conducted are found in major urban areas such as Kathmandu; therefore, people residing in rural areas have to travel for 8 to 12 hours and pay NPR 8,000 to NPR 15,000. This signifies that the cost is more than the earning made by an individual in a month; hence, many brain diseases go undiagnosed until they become severe.

However, recent developments in artificial intelligence and especially deep learning show that this problem may be solved soon. In particular, based on current research, artificial intelligence can analyze MRI images just as efficiently as doctors (Berghout, 2024; Prasanthi & Neelima, 2024). Unfortunately, translation of these systems into practical tools for resource limited clinical environments has been slow and the literature suggests very few published systems have been designed with places like Nepal specifically in mind.

NeuroScan Nepal is an effort to create something that bridges this gap in a concrete and pragmatic way. The system will examine the uploaded MRI images and classify the image either as abnormal or normal. In case the system detects an abnormal image, it will then attempt to understand the type of anomaly found in the image. A Retrieval-Augmented Generation (RAG) component will generate advice on the subject that can be applied to the context. A bilingual chatbot using natural language processing techniques will allow users to ask queries pertaining to the analysis in both English and Nepali languages. The data for the model would be taken from MRI scans that can be accessed from the Grande International Hospital, Kathmandu. Access to real Nepali clinical imaging data.

1.1 Problem Statement

Healthcare workers in resource-limited settings across Nepal lack access to affordable, easy-to-use tools that can accurately identify brain abnormalities from MRI scans and provide meaningful clinical guidance in local languages. This contributes to delayed diagnoses, higher risks for patients, and a deeply unequal distribution of neurological healthcare across the country.

2 Review of Existing Knowledge

2.1 Early Techniques in AI-Based Medical Imaging

Research on automated brain abnormality detection began with classical machine learning methods such as support vector machines and random forests applied to manually engineered image features. However, despite the initial success they enjoyed, such approaches had some difficulty when it came to generalisation because of which they could not perform reliably under varying imaging circumstances and populations, thus making them impractical in clinical settings (Bairagi et al., 2023; Zulfiqar et al., 2023). The field of medical image analysis and classification has been revolutionised by deep learning methods, especially convolutional neural networks. Methods using deep learning could use raw images themselves to discover useful features in them and could distinguish different tumour types from normal tissue in MRIs with accuracies comparable to humans, even in controlled conditions (Berghout, 2024; Prasanthi and Neelima, 2024). The process was also faster through transfer learning, where ResNet and EfficientNet took advantage of feature extraction from large databases to obtain high accuracy, even with small medical data (Zulfiqar et al., 2023; Tenghongsakul et al., 2023). Vision Transformers can comprehend the global spatial context of an MRI scan (Krishnan et al., 2024), although these are computationally expensive.

2.2 Explainability and the Black Box Problem

Deep learning models, even with good performance, have been heavily criticised for being difficult to understand. Early diagnostic systems based on CNNs served as black boxes, producing classification outputs with no explanation of the reasoning behind the decision, which naturally made medical professionals reluctant to trust or adopt them (Li and Dib, 2024). This is a valid concern in a clinical setting where the consequences of a wrong prediction can be serious. However, there are some AI approaches that could potentially be used to solve the stated problem. For example, Grad-CAM provides heat maps and points to the areas within the MRI picture that have been important for forming the prediction, whereas approaches such as LIME and SHAP rely on explaining the features behind predictions (Afnaan et al., 2025; Rahman et al., 2025). This method makes it possible to validate the model's decisions. That said, explainability techniques are studied extensively but are still rarely integrated into fully deployable clinical systems, which is a gap I intend to address in NeuroScan Nepal through Grad-CAM visualisation.

2.3 Deployment in Low-Resource Settings

The most significant problem with the current literature is its neglect of practicality, especially in settings with scarce resources. Most studies have relied on an algorithmic framework created within optimal research environments where the use of high-end GPUs and cloud computing are possible, something which is not feasible in healthcare facilities in Nepal (Neamah et al., 2023). Strategies such as model quantisation and pruning have been proposed to reduce the computational demands of deep learning models, and FPGA-based offline deployment has been demonstrated in at least one study (Lata et al., 2025). However, these approaches are rarely brought together into complete systems that can function fully offline while also providing meaningful clinical support for non-specialist users.

2.4 Dataset Limitations

Moreover, the problem of data set quality and data set variety remains an important concern. The problem lies in that many public MRI data sets often show an imbalanced class, an inconsistency in the imaging format, and no representation of patients from other countries, such as Nepal (Bairagi et al., 2023; Caragliano et al., 2025). As a consequence, the applicability and capability of the trained models to generalise will be hindered when employed in real clinical settings. Also, model evaluations are generally done through technical factors like accuracy and F1-score, ignoring usability and clinical relevance (Neamah et al., 2023). The gap between a good research model and a tool that can be used by a healthcare worker in a district hospital is still wide.

2.5 Critical Analysis and Identification of Gaps

A review of the current literature shows that there are three clear gaps. To start with, no system exists that would allow the identification of abnormalities in the brain through its full application in situations when resources such as specialists and technology are scarce in places such as Nepal (Neamah et al., 2023; Lata et al., 2025). Controlled settings have resulted in high accuracy for existing systems, but they were not intended for practical use in clinics. The second gap is the absence of integrated clinical guidance for the non-specialist user. Current systems usually provide a classification label and perhaps a confidence score but nothing else to help a junior doctor or community health volunteer to understand what this result means, what they should do about it or where they can refer the patient. This is a significant limitation when the intended users may not have the specialist training to interpret a bare classification output meaningfully (Caragliano et al., 2025). The third gap is the lack of attention given to issues such as usability, communication with the patients, and automated reporting in the published studies, coupled with the lack of any brain abnormality diagnosis AI system that produces output in both English and Nepali languages. The importance of this feature lies in the fact that Nepali is the main language spoken by health-care workers and patients, and thus this omission is vital.

2.6 Project Justification

All the above issues will be sorted out by NeuroScan Nepal. First off, employing deep learning architectures that require minimal computational power allows developing the solution right from the start with the focus on practical application, rather than doing this retroactively. Secondly, incorporating the functionality of RAG and creating a bilingual NLP chatbot will provide medical guidance and support of the conversational nature that no current study addresses in terms of practical implementation. Thirdly, a structured usability test involving at least 20 participants will produce results regarding usefulness that is almost entirely lacking in existing research. Lastly, anonymized MRI scans of patients obtained from Grande International Hospital, Kathmandu will make sure that the model is trained on relevant data concerning Nepal only, a feat that cannot be accomplished by any public domain benchmark set.

3 Project Aims, Objectives, and Scope

3.1 Project Aim

This project aims to design, develop, and evaluate NeuroScan Nepal which an intelli-gent AI-powered brain abnormality detection and conversational advisory system that uses deep learning to classify MRI brain scan images as normal or abnormal, a RAG module to generate contextually relevant medical guidance, and a bilingual NLP chatbot to support healthcare workers and patients in both English and Nepali which with the goal of improving early detection and reducing diagnostic inequality in resource-limited healthcare settings in Nepal.

3.2 Project Objectives

1. The objective is to apply Transfer Learning to train the CNN model on patient's anonymous MRI images available from Grande International Hospital, Kathmandu, with an accuracy of at least 90% in distinguishing normal and abnormal brain appearances in 4 months.

2. In order to incorporate and implement a RAG model based on a carefully selected set of medical data comprising a minimum of 50 medical documents, ensuring the system provides accurate advice for any abnormality classifications within 5 months.

3. To build a bilingual NLP Chatbot for conversational purposes in English and Nepali, with the ability to give context-based answers to the queries of patients as well as medical personnel within 6 months.

4. To create a Nepali healthcare database which includes at least 10 neurology hospitals, healthcare programs by the government and associated costs, within 5 months.

5. To deploy a user-friendly Flask web application integrating MRI upload, detection results, RAG advisory output, chatbot support, Grad-CAM visualisation, and downloadable PDF report generation within 7 months.

6. For conducting a usability test with a minimum of 20 users in which there is a user satisfaction of at least 80% with documentation of system weaknesses and future scope within 8 months.

3.3 Project Scope

The system will be able to classify axial view MRI brain scan images in JPG or PNG format into either normal or abnormal, with further characterization of the abnormality in cases where data is available. The primary dataset will consist of anonymised clinical MRI records from Grande International Hospital, Kathmandu. Publicly available datasets will be used as supplementary data if needed. The RAG knowledge base will cover brain abnormalities and Nepal-specific healthcare information. The chatbot will support English and Nepali. The system will not have integration with hospital information systems; will not provide a definitive clinical diagnosis; and will carry compulsory medical disclaimers in all outputs provided by it. There will be no personal patient information involved. Medical device certification is not covered under this project.

4 Project Design

4.1 Design and Methods

NeuroScan Nepal will be built in an iterative agile development process with clear phases and milestone reviews at the end of the development of each major component. This approach is appropriate because the system components are tightly coupled, changes in the model can impact the advisory system, and changes in the system interface can impact the output of usability evaluation. In iterative development, the whole project cycle is a chance for continual improvement. The brain MRI images used are anonymized data from patients at Grande.

Through a formal research collaboration, International Hospital, Kathmandu, was generated. These data sets contain variations in the data set (such as contrast, positioning, artefacts, patient complexity etc.) which are not present in public databases. In Nepal they perform normal and abnormal practices in their regular clinical practice. There will be approximately 500 images labeled. If necessary, publicly available MRI datasets will be used as additional data. The image pre-processing includes Contrast Limited Adaptive Histogram Equalization (CLAHE), resizing the image to 224x224 pixels, and normalization of the channels as well as data augmentation operations: rotation, zoom, horizon flip and brightness adjustment. The techniques will be helpful to avoid overfitting, particularly with the relatively small set of data. We will train and compare a custom CNN baseline model, a VGG16 transfer learning model and an EfficientNetB0 transfer learning model. We will test the performance of the model using accuracy, precision, recall, f1 score, AUC-ROC, and cross validation to make our model robust.

Langchain will be the orchestration framework, FAISS the vector database, and HuggingFace sentence transformers to generate embeddings. A locally hosted large language model (Llama 3 or Mistral as per availability of hardware) will be trained to generate responses based on retrieved medical context. All these will be combined in a single system, the Flask web application, which will include the upload of MRI, inference of the model, output of the advice in PDF using RAG, and interaction with the chatbot.

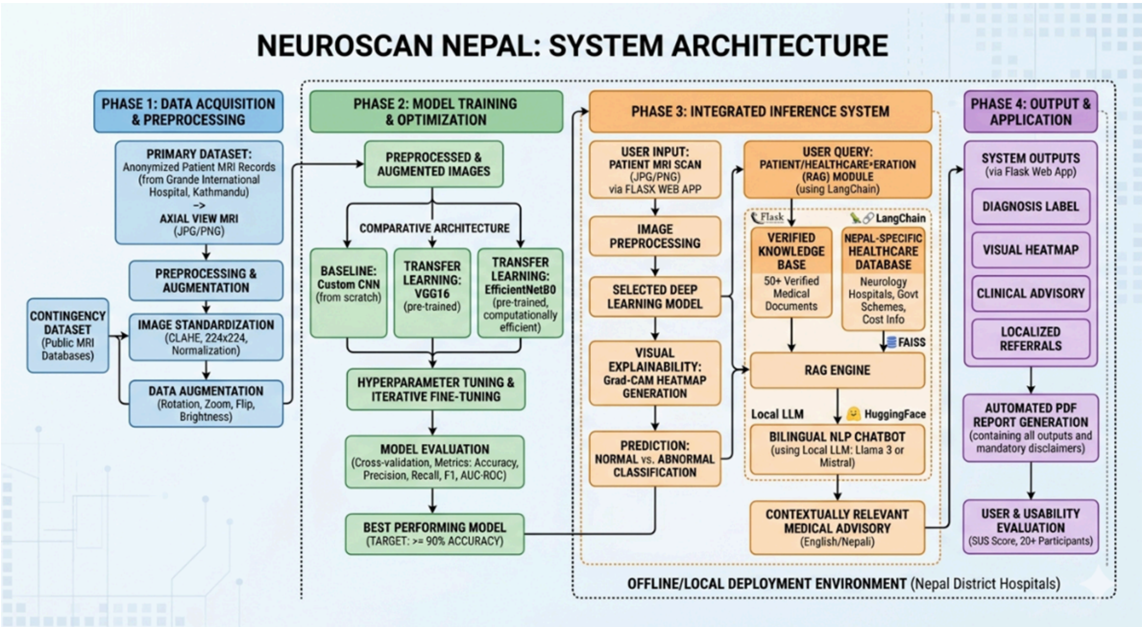

Figure 4.1: NeuroScan Nepal System Architecture and Workflow

4.2 Justification of Methods

Transfer Learning has been chosen because it is better compared to Training from Scratch as Transfer Learning always beats in small medical image datasets because of its learnt feature representation. The design of the experiment would be better to compare EfficientNetB0, VGG16 and a custom CNN, since it provides a balance between the modern architectures and a base model. EfficientNetB0 is chosen because of its performance & computation efficiency.

LangChain and FAISS are chosen because they have proven to be effective in the Python programming environment. For the sake of effective support in Python programming environment, we put LangChain and FAISS in. We're running a local LLM to serve offline, save money, and ensure privacy of medical information. Flask is chosen due to its lightweight and machine learning workflow compatibility. The System Usability Scale (SUS) will be used as it is a well established instrument for gauging system usability.

4.3 Project Timeline

Task / Phase	M1	M2	M3	M4	M5	M6
Literature Review & Research						
Dataset Collection (Grande Hospital)						
Data Preprocessing & Augmentation						
Model Training & Architecture Comparison						
RAG Module Development						
Chatbot Implementation						
Flask Web Application Development						
System Integration & Testing						
Usability Evaluation						
Final Report Writing						

Figure 4.3: Project Timeline (M1–M6=Months 1-6; shaded = active phases)

5 Feasibility, Risks, and Ethical Considertations

5.1 Feasibility

The project is deemed realistic in terms of the time and resources available for the final year undergraduate project. One practical advantage is that the main data set has already been partially acquired in the form of anonymised MRI scan data from Grande International Hospital, Kathmandu, acquired as part of a formal clinical collaboration. It also ensures that data collection is not solely reliant on any future approval since the process has commenced even before the completion of ethical documentation. All core tools required for implementation, including TensorFlow, Keras, LangChain, FAISS, Flask, and Hugging Face Trans-formers, are open-source and freely available, with no licensing costs involved. A strong foundation has been built for Python and Machine Learning in the degree course and in the student's personal work and research in Retrieval-Augmented Generation (RAG) systems and LangChain. Furthermore, public bench MRI datasets have been recognized as a contingency to help avoid potential delays in formal data transfer.

5.2 Risk Analysis and Mitigation

Potential Risk	Likelihood	Impact	Mitigation Strategy
Delay in formal data transfer from Grande International Hospital beyond CD already received	Low	High	CD already in possession. Formal data sharing agreement and BCU ethics forms in active preparation. Publicly available MRI datasets identified as immediate contingency.
Model accuracy falls below 90% target on the clinical dataset	Medium	High	Three architectures compared with iterative hyperparameter tuning, augmentation, and fine-tuning. Dataset will be supplemented with public MRI data if accuracy remains insufficient. Supervisor consulted at each milestone.
RAG and LangChain imple-mentation proves more com-plex than expected	High	Medium	Dedicated learning time allocated in Month 5. Extensive documenta-tion and community tutorials avail-able. Supervisor guidance sought promptly for any technical blockers.
Difficulty recruiting 20+ us-ability testing participants	Medium	Medium	Recruitment begins early in Month 7 using BCU student network and Nepal healthcare contacts. Remote online testing included to broaden participation.
Loss of data or project files due to hardware failure	Low	High	All files backed up weekly to an en-crypted external drive and private GitHub repository throughout the project.

Table 1: Risk Assessment and Mitigation Strategy

5.3 Ethical Considerations

Ethical considerations have been taken into account in the whole process of the project owing to the usage of the sensitive personal data of a medical nature. The BCU Student Project Ethics Declaration Form has been initiated and will be finalized before submission by both the student and the supervisor. The project uses the MRI record of patients acquired within the hospital environment; thus, all necessary ethics procedures will be completed in line with BCU CDT guidelines. A comprehensive set of documents related to ethics, namely an Access Request Letter, a Consent Form, and a Data Sharing Agreement, has been produced in line with institutional templates, and is expected to be approved by Grande International Hospital before utilizing the data. Although a preliminary set of anonymised MRI scans has been received for review purposes, none of the data will be used for training until full ethical approval has been granted.

All patients' personal information will be carefully anonymized by qualified radiologists before it is provided to the hospitals to remove personal data such as names, birth dates and ID numbers. The de-identified data set will be stored on an encrypted computer, which can only be accessed by the researcher and supervisor with the use of passwords. Data will be gathered solely for research and will be destroyed after the project is finished and/or the student examination.

All the subjects will be provided with an information sheet and informed consent before participation for usability evaluation. Participants will be free to withdraw at any time and they will not be penalized for their choice to participate. Limited demographic information will be gathered for analysis. The system will prominently display disclaimers that all outputs are medical in nature, disclaim that the system provides medical support but not a replacement for professional medical judgment, and will have required medical disclaimers on all outputs.

5.3.1 Summary Table for Ethical Consideration

Ethical Area	**Concern**	**Mitigation Strategy**
Patient Data Privacy	Use of sensitive MRI medical data	Full anonymisation under radiologist supervision; removal of all identifiers; encrypted storage with restricted ac-cess; data deleted after project comple-tion.
Ethics Approval Com-pliance	Use of hospital data prior to approval	No training performed until full BCU ethics approval and signed data-sharing agreement are completed.
Informed Consent	Use of participant data in usability test-ing	Participants provided with information sheet and consent form; participation voluntary with right to withdraw at any time.
Data Security	Risk of unauthorized access or data loss	Encrypted, password-protected storage with weekly backups to secure external drive and private GitHub repository.
Medical Misinterpreta-tion Risk	Misuse of system out-puts by users	System includes clear medical dis-claimers; chatbot directs users to pro-fessional medical consultation; tool de-signed as decision-support only.

Table 2: Ethical Considerations and Mitigation Strategy

6 Bibliography

- Afnaan, K., Arunbalaji, C.G., Singh, T., Kumar, R. and Naik, G.R. (2025) ’Boosting brain tumor detection with an optimized ResNet and explainability via Grad-CAM and LIME’, *Brain Informatics*, 12(1), p. 33. doi: 10.1186/s40708-025-00242-x.
- Alokozay, E., Haider, E., Waseem, N. et al. (2025) ’Addressing global dis-parities in neurosurgical workforce and access to care’, *Chinese Neurosurgical Journal*, 11, p. 30. doi: 10.1186/s41016-025-00393-2.
- Bairagi, V.K., Gumaste, P.P., Rajput, S.H. and Chethan, K.S. (2023) ’Au-tomatic brain tumor detection using CNN transfer learning approach’, *Med-ical and Biological Engineering and Computing*, 61(7), pp. 1821–1836. doi: 10.1007/s11517-023-02823-2.
- Berghout, T. (2024) ’The neural frontier of future medical imaging: A review of deep learning for brain tumor detection’, *Journal of Imaging*, 11(1), p. 2. doi: 10.3390/jimaging11010002.
- Caragliano, A.N. et al. (2025) ’Exploring the limitations of virtual contrast prediction in brain tumor imaging’, *NMR in Biomedicine*, 38(6), p. e70062. doi: 10.1002/nbm.70062.
- Dixit, A. and Thakur, M.K. (2025) ’3W-MultiHier: A three way multi-hierarchical model enabled deep learning for brain tumor classification in MRI scans’, *Ad-vanced Theory and Simulations*, 8(3), p. 2400752. doi: 10.1002/adts.202400752.
- Krishnan, P.T. et al. (2024) ’Enhancing brain tumor detection in MRI with a rotation invariant Vision Transformer’, *Frontiers in Neuroinformatics*, 18, p. 1414925. doi: 10.3389/fninf.2024.1414925.
- Lata, K. et al. (2025) ’Privacy-preserving brain tumor detection using FPGA-accelerated deep learning on Kria KV260 for smart healthcare’, *Computer Methods and Programs in Biomedicine Update*, p. 100205. doi: 10.1016/j.cmpbup.2025.10020
- Lewis, P. et al. (2020) ’Retrieval-augmented generation for knowledge-intensive NLP tasks’, *Advances in Neural Information Processing Systems*, 33, pp. 9459–9474. Available at: https://proceedings.neurips.cc/paper/2020 (Accessed: 9 May 2026).
- Li, Z. and Dib, O. (2024) ’Empowering brain tumor diagnosis through ex-plainable deep learning’, *Machine Learning and Knowledge Extraction*, 6(4), pp. 2248–2281. doi: 10.3390/make6040110.
- Neamah, K., Mohamed, F., Adnan, M.M., Saba, T., Bahaj, S.A., Kadhim, K.A. and Khan, A.R. (2023) ’Brain tumor classification and detection based DL models: A systematic review’, *IEEE Access*, 12, pp. 25172–25142. doi:

10.1109/ACCESS.2023.3340657.

- Prasanthi, T.L. and Neelima, N. (2024) 'Improvement of brain tumor catego-rization using deep learning: A comprehensive investigation and comparative analysis', *Procedia Computer Science*, 233, pp. 703–712. doi: 10.1016/j.procs.2024.03.234.
- Rahman, A. et al. (2025) 'Enhanced MRI brain tumor detection using deep learning in conjunction with explainable AI SHAP based diverse and multi feature analysis', *Scientific Reports*, 15(1), p. 29411. doi: 10.1038/s41598-025-29411-x.
- Shrestha, P. (2024) 'Neurosurgical workforce of Nepal', *World Neurosurgery*, 189, pp. 161–165. doi: 10.1016/j.wneu.2024.06.101.
- Srinivas, C. et al. (2022) 'Deep transfer learning approaches in performance analysis of brain tumor classification using MRI images', *Journal of Healthcare Engineering*, 2022, p. 3264367. doi: 10.1155/2022/3264367.
- Tenghongsakul, K. et al. (2023) 'Deep transfer learning for brain tumor detec-tion based on MRI images', *Journal of Physics: Conference Series*, 2497(1), p. 012015. doi: 10.1088/1742-6596/2497/1/012015.
- Zulfiqar, F., Bajwa, U.I. and Mehmood, Y. (2023) 'Multi-class classification of brain tumor types from MR images using EfficientNets', *Biomedical Signal Processing and Control*, 84, p. 104777. doi: 10.1016/j.bspc.2022.04.777.